



澳門科技大學
MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

BE202 EX4 Final Project

Glove Mouse

Xie, Wen-hao 1230021133

Shi, Zhi-bo 1230017906

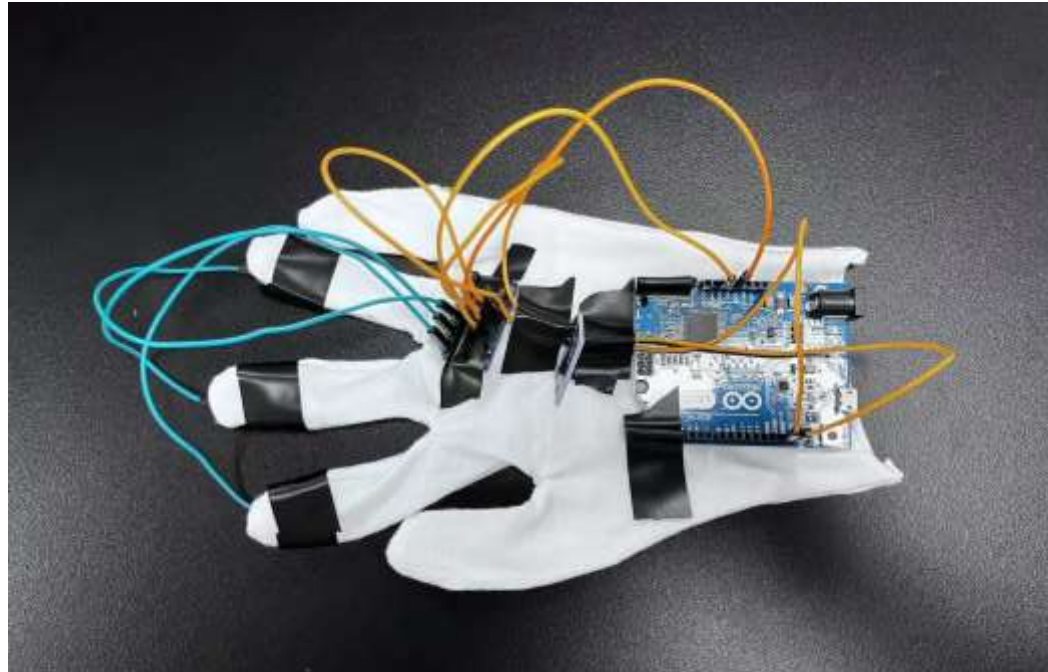
Background

- The development of a wearable gesture-based glove mouse is essential for enhancing user interaction and accessibility in computing. This device enables cursor control through natural hand movements, offering a more intuitive and interesting experience.

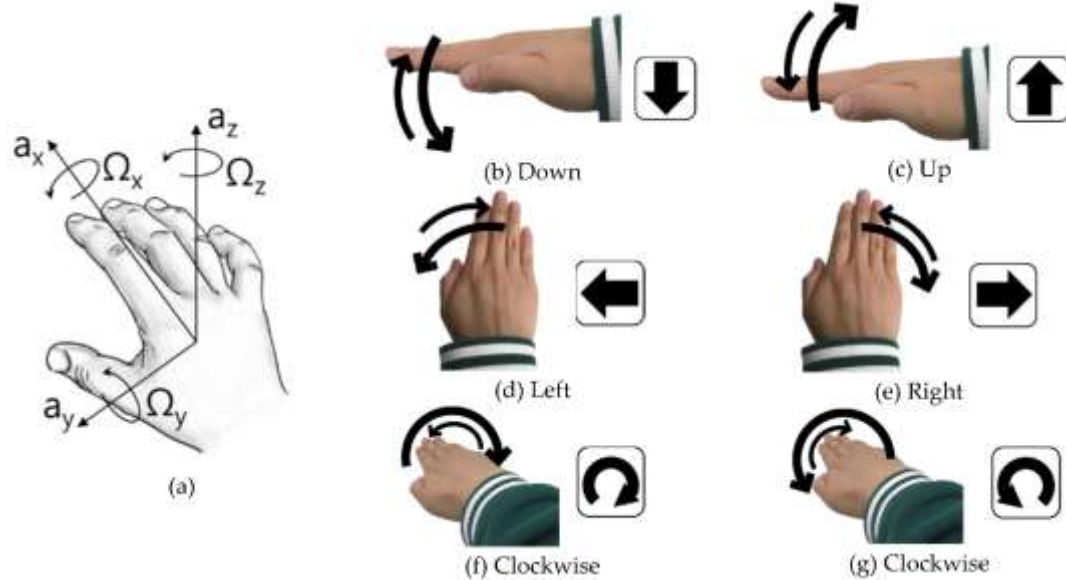


Description

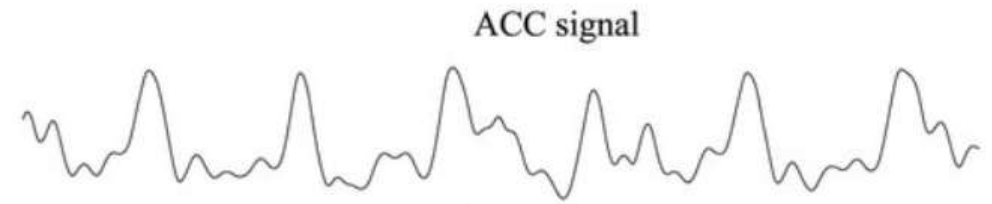
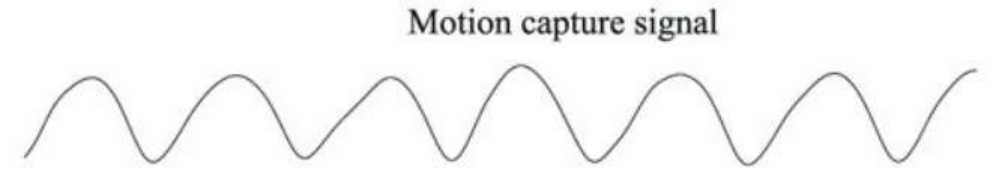
In this project, we demo a glove mouse using an Arduino board. It transforms hand motions into cursor movements. It sends commands to the computer via connection. Buttons can mimic mouse clicks. This glove mouse provides a new way for human-computer interaction.



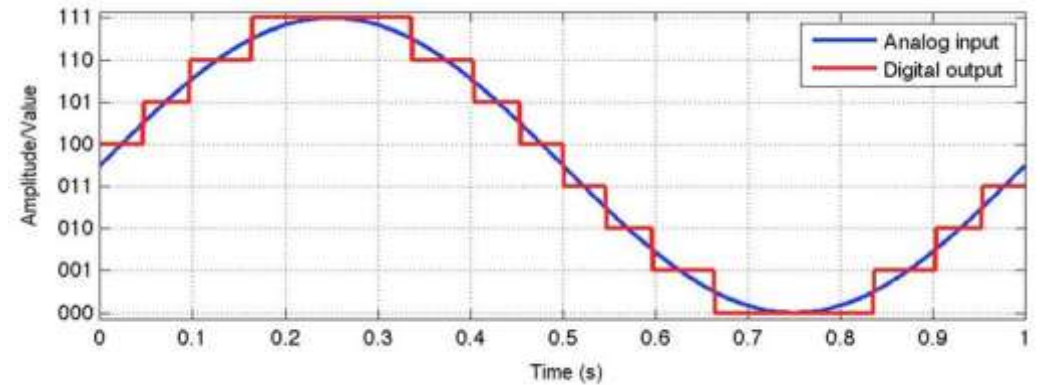
Principle and Scheme



Motion
signal

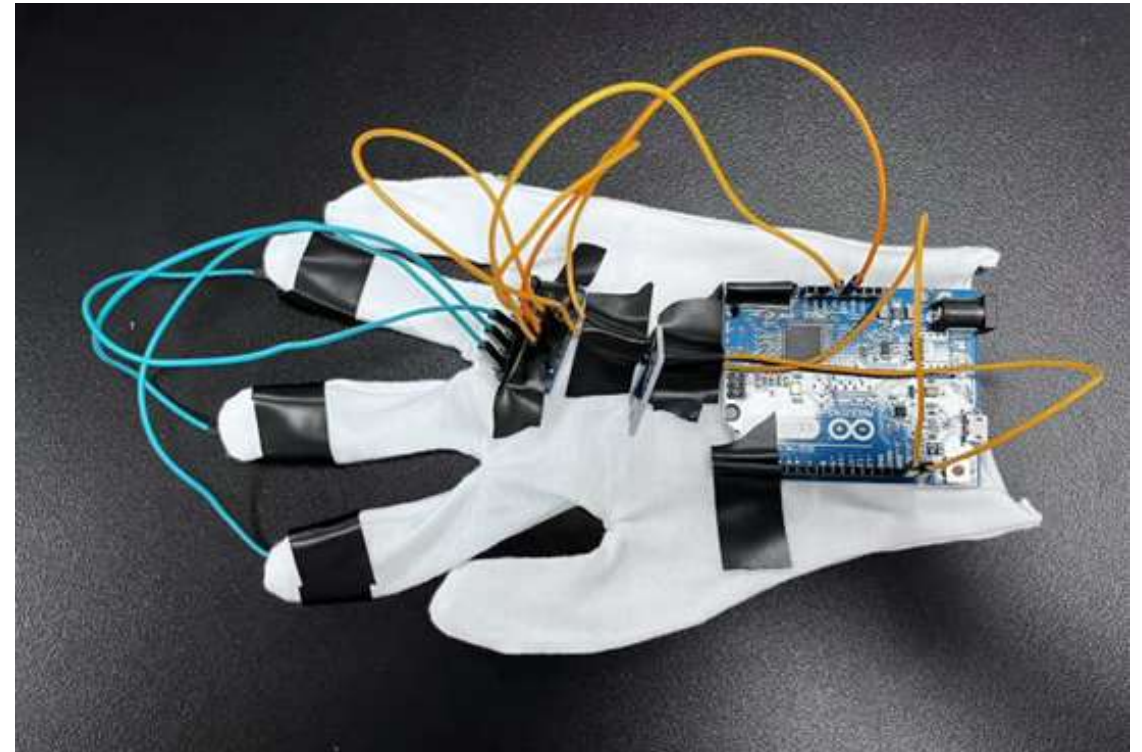
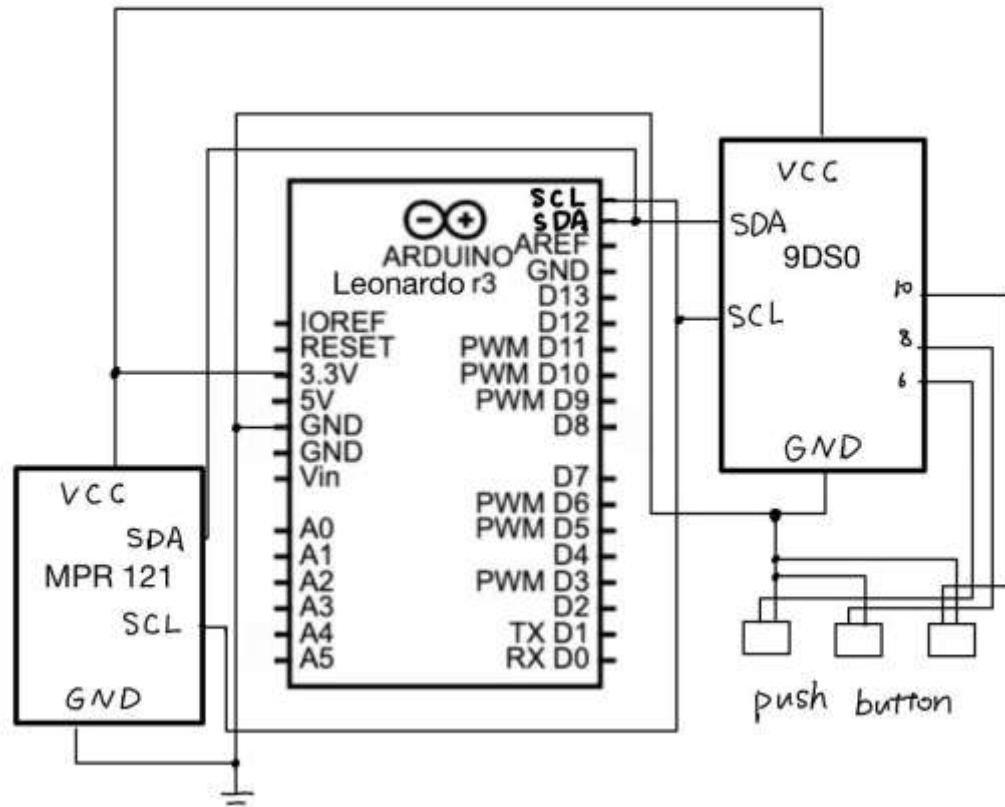


Analog
to
Digital



Digital signal control cursor movement

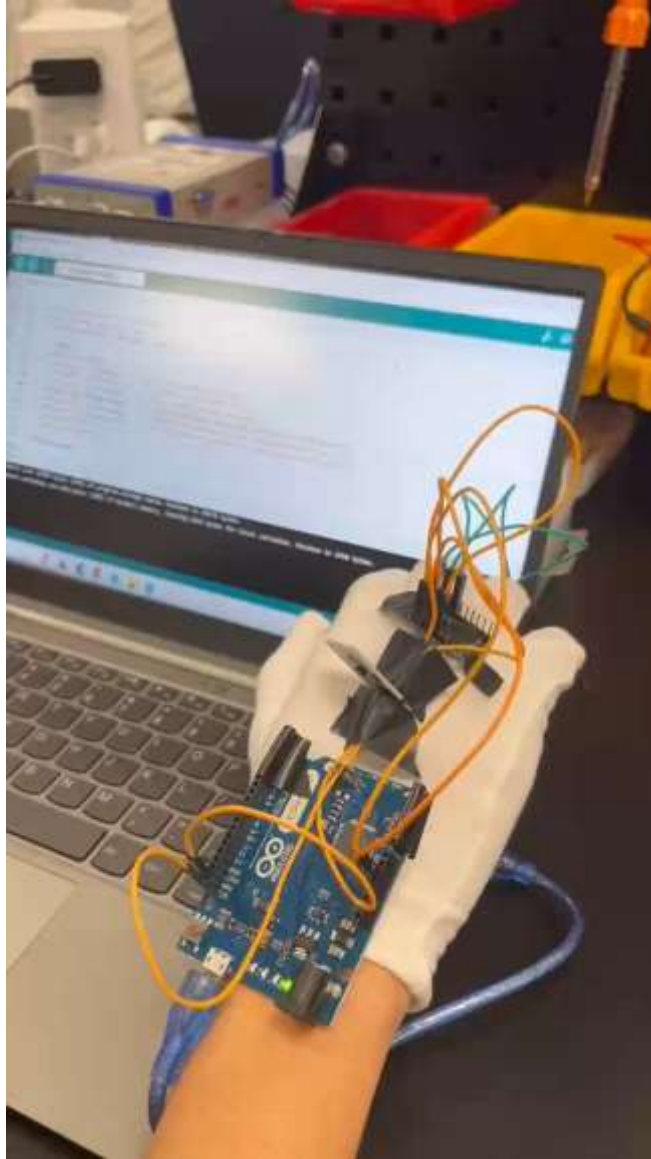
Circuit Diagram of Glove Mouse



Program Scheme

1. The sensor obtains accelerometer, magnetometer, gyroscope and current touch status data.
2. The accelerometer extracts X and Y axis data, calculates the cursor movement according to the acceleration data, and controls the cursor movement.
3. The touch sensor processes data, presses and releases the mouse button according to the change of touch status, and controls the mouse button.

Demonstration of Glove Mouse



Authors and Reference

Authors

Shi Zhi-bo

Xie Wen-hao

Reference

- Han, H., & Yoon, S. W. (2019). Gyroscope-Based Continuous Human Hand Gesture Recognition for Multi-Modal Wearable Input Device for Human Machine Interaction. Sensors, 19(11), 2562. <https://doi.org/10.3390/s19112562>
- Paul, W. (n.d.). Mouse & Rat: Technical Explanation. Retrieved from <https://will-paul.com/posts/mouse-rat-technical-explanation/>
- Behera, A., P., A. K., Badiger, A. S., Hemanth, & Jeyaseelan, V. (2023). WEARABLE GLOVE MOUSE. IJARIE, 9(3), 570-580. Retrieved from https://ijarie.com/AdminUploadPdf/Wearable_Glove_Mouse_ijarie20100.pdf?srsltid=AfmBOoqqY0Jc4KH_dVGSD8vFF1M WG8tatIMiRRX0dAuJTo04ZcE-5Qrn
- Arduino Mouse Control Glove. (n.d.). Retrieved from <https://www.instructables.com/Arduino-Mouse-Control-Glove/>
- Gesture Mouse - Controlling Computer With a Glove. (n.d.). Retrieved from <https://www.instructables.com/Gesture-Mouse-Controlling-Computer-With-a-Glove/>



澳門科技大學

MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thank you for your attention!



澳門科技大學
MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY